Let $x \neq 2m\pi$ ($m \in \mathbb{Z}$) be a real number. Observe that

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\cdot\sum_{k=0}^{n-1} e^{ikx} = e^{ix}\cdot\frac{1-e^{inx}}{1-e^{ix}} = e^{ix}\cdot\frac{e^{\frac{1}{2}inx}}{e^{\frac{1}{2}ix}}\cdot\frac{e^{\frac{1}{2}inx}-e^{-\frac{1}{2}inx}}{e^{\frac{1}{2}ix}-e^{-\frac{1}{2}ix}}$$

$$= e^{\frac{ix}{2}\cdot(n+1)}\cdot\frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}$$

Since $e^{ikx} = \cos kx + i\cdot\sin kx$ and

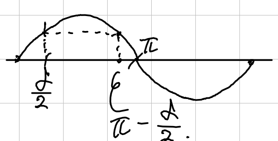
$$e^{\frac{ix}{2}(n+1)} = \cos\frac{(n+1)}{2}x + i\cdot\sin\frac{(n+1)}{2}x$$

So therefore

$$\sum_{k=1}^{n}\cos kx = \frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}\cdot\cos\frac{n+1}{2}\cdot x \quad \text{and} \quad \sum_{k=1}^{n}\sin kx = \frac{\sin\frac{nx}{2}}{\sin\frac{x}{2}}\cdot\sin\frac{n+1}{2}\cdot x$$

Now, for fixed $0<\delta<\pi$, given $x\in[\delta, 2\pi-\delta]$,

$$\left|\sum_{k=1}^{n}\cos kx\right|,\ \left|\sum_{k=1}^{n}\sin kx\right| \leq \left|\frac{1}{\sin\frac{x}{2}}\right| \leq \left|\frac{1}{\sin\frac{\delta}{2}}\right|$$



Consequently, we obtain a Theorem;

Thm. Suppose that $\{a_n\}$ converges to 0, and $\sum|a_n - a_{n-1}|$ converges.
Given $\delta\in(0,\pi)$, $\sum_{n=1}^{\infty} a_n\cdot\sin nx$ and $\sum_{n=1}^{\infty} a_n\cdot\cos nx$ converges uniformly on $[\delta, 2\pi-\delta]$.

Proof. Since the observation above,

$$\left|\sum_{n=1}^{m}\sin nx\right| \leq \left|\frac{1}{\sin\frac{\delta}{2}}\right| \quad \text{for all } \delta\leq x\leq 2\pi-\delta.$$

for all $m\in\mathbb{N}$, so $\sum\sin nx$ is bounded uniformly. So, it satisfies the condition for the Diriclet-Dedekind Test, the given series converges uniformly on $[\delta, 2\pi-\delta]$.

Thm. Let $\{a_n\}$ be a monotone-decreasing real-numbers sequence, converging to 0. Then, $\sum_{n=1}^{\infty} a_n \cdot \sin nx$ converges uniformly on $\mathbb{R}$ if and only if $\lim_{n\to\infty} n \cdot a_n = 0$.

Proof. Suppose that $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$.
By the Cauchy criterion, given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$n > m \geq N \Rightarrow \left| \sum_{k=m}^{n} a_k \cdot \sin kx \right| < \varepsilon \quad \text{for all } x \in \mathbb{R}.$$

Now, for any $n \in \mathbb{N}$, let $m \in \mathbb{N}$ be the smallest integer bigger than $\frac{n}{2}$.
Then, for all $k = m, m+1, \cdots, n$, $\frac{1}{2} \geq \frac{k}{2n} \geq \frac{m}{2n} \geq \frac{1}{4}$. Therefore,

$$\sum_{k=m}^{n} a_n \cdot \sin \frac{k}{2n} \cdot \pi \overset{(1)}{\geq} \sum_{k=m}^{n} a_n \cdot \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \cdot \sum_{k=m}^{n} a_n \overset{(2)}{\geq} \frac{1}{\sqrt{2}} \cdot \frac{n}{2} \cdot a_n$$

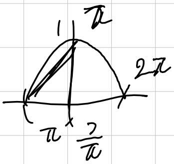
The (1) is given by the fact that $\sin x$ increase on $[0, \frac{\pi}{2}]$,
The (2) is given by the fact that $\{a_n\}$ decrease monotonically.
Now, the first term in the inequality tends to zero, so is $n \cdot a_n$.
Conversely, suppose that $n a_n \to 0$ as $n \to \infty$. For each $n \in \mathbb{N}$, define $\varepsilon_n := \sup_{k \geq n} k \cdot a_k$.
Let $x \in [0, \pi]$ be fixed. Then, there exists a positive integer $N \in \mathbb{N}$ such that

$$\frac{\pi}{N+1} < x < \frac{\pi}{N}$$

So, $$\left| \sum_{k=n}^{n+N-1} a_k \cdot \sin kx \right| \leq \sum_{k=n}^{n+N-1} a_k \cdot |\sin kx| \leq \sum_{k=n}^{n+N-1} a_k \cdot kx \leq N \cdot \varepsilon_n \cdot x < \pi \cdot \varepsilon_n$$

Meanwhile, let $S_n = \sum_{k=1}^{n} \sin kx$. Then, given $M > n+N$,

$$\begin{aligned}
\left| \sum_{k=n+N}^{M} a_k \cdot \sin kx \right| &= \left| \sum_{k=n+N}^{M} a_k \cdot (S_k - S_{k-1}) \right| \\
&= \left| \sum_{k=n+N}^{M} a_k \cdot S_k - \sum_{k=n+N}^{M} a_k \cdot S_{k-1} \right| \\
&= \left| \sum_{k=n+N}^{M} [a_k - a_{k+1}] \cdot S_k + a_{M+1} \cdot S_M - a_{n+N} \cdot S_{n+N-1} \right| \\
&\leq \left| \frac{1}{\sin \frac{x}{2}} \right| \cdot 2a_{n+N} \leq \frac{\pi}{x} \cdot 2a_{n+N} < 2(N+1) \cdot a_{n+N} < 2 \cdot (N+n) \cdot a_{n+N} < 2\varepsilon_n
\end{aligned}$$



Consequently, for each $n \in \mathbb{N}$, $\left| \sum_{k=n}^{\infty} a_k \cdot \sin kx \right| < (2+\pi) \cdot \varepsilon_n \to 0$ as $n \to \infty$

So the given series converges uniformly on $[0, \pi]$, the periodicity gives convergense on $\mathbb{R}$.